Enforcing Hidden-State Policies in Native Private Aggregation

Abstract—**A managed fleet task can aggregate reports from a complete predeclared cohort while keeping client state hidden from the admission authority. A malicious report that moves a contribution to another cell, or the omission of an accepted report, can distort the resulting heatmap even when every aggregated input is individually valid. These attacks change a payload relative to an authorized claim. A fabricated but policy-conformant claimed trajectory remains outside the physical-truth guarantee. The aggregate therefore needs to reflect the transitions authorized against each client's hidden history. Stateful History-to-Payload Closure (SHTPC) connects those transitions to the outputs retained by Prio3 and to the complete cohort exposed at release. Its masked moment uses native one-hot validity to bind the authorized cell without reproducing native seed expansion in the application circuit. The standalone binding relation has 5,779 constraints at each evaluated histogram size, while native processing still scans the vector. Atomic admission advances the history, and durable reservation fixes the cohort before an honest aggregate share becomes decryptable. With correctly executing services, the integrity analysis covers accepted history across windows and first exposure. A separate privacy analysis covers one complete cohort per task and key against an active aggregator colluding with the Collector. It compares legal histories with common public metadata and equal declared cohort sums. Janus tests reject native-valid mismatches, block a distinguishing subset exposure and recover exact 100-member outputs after withheld deliveries.**

1. Introduction

A fleet operator uses a heatmap to identify congestion and allocate vehicles, while drivers keep their locations private. Private location statistics and accountable vehicular services motivate this combination of confidentiality and admissible contributions [1–3]. If a participant redirects a report from its authorized cell to a target cell, the target count rises and the original count falls. A dispatcher can then act on a hotspot that the authorized reports do not support. Omitting an accepted report can similarly hide demand. These errors concern the meaning and membership of the aggregate, even when the aggregation arithmetic is correct.

In a controlled offline replay of 100 selected T-Drive histories, redirecting 20 reports to the clean modal cell raises its count from 3 to 22–23 across three fixed actor permutations. An exact-set release contract then withholds the window if substituted contributions remain unresolved. Section 5.2 examines this integrity and availability tradeoff with the selection losses of the real trajectories.

Periodic reporting motivates complete cohorts when a missing accepted report can distort a decision. The integrity contract supports continued accepted history across windows, including after abort. The privacy theorem covers one visible complete cohort per task and key, not the composition of multiple outputs, even from nonoverlapping windows. An admission authority authorizes transitions and records quotas without reading client coordinates. Two aggregation roles process secret shares, and a Collector consumes statistics for a fixed cohort. DAP with Prio3 [4–6] provides native validation and private aggregation. The managed services enforce the integrity contract, while clients and transport may be malicious. The application needs the authorized transitions, retained contributions and release membership to describe the same accepted history.

Existing protocols validate private inputs [7–10], enforce inclusion or verify aggregates [11–13], and maintain private aggregate state [14]. The application here needs these interfaces to refer to one accepted history. Consider a proof that authorizes cell p and a native-valid report that encodes cell $q \neq p$. Both objects pass their respective checks even when they share a report identifier and a signed body. The identifier authenticates their association, not their semantic equality. A hiding commitment likewise binds its committed value without proving that this value determines the native output. The missing relation connects the hidden witness to the output actually retained for aggregation. A generic commitment scheme that proves this equality would close this particular gap.

State freshness and release membership create two further interfaces. A proof for predecessor c remains valid after a competing successor commits, so proof verification does not authorize the current transition. Separately, an accepted report can be dropped on both receiver channels. The receivers then agree on a valid but incomplete batch, as Figure 1 shows. These failures break the same connection between the history authorized by the application, the values aggregated and the cohort exposed to the Collector.

Completeness also has a privacy deadline. Two legal histories can have identical complete heatmaps but different subset aggregates. A final publication check can refuse an incomplete cohort, but it cannot retract an aggregate share that already exposed the subset. Accepted-history closure therefore connects each admitted claim to its consumed payload and atomic state transition. It also reconciles the complete eligible set before the first honest share becomes decryptable.

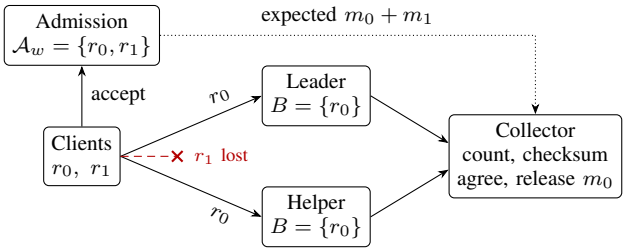


Figure 1. Both aggregators can agree on a batch that omits an accepted report.

Stateful History-to-Payload Closure (SHTPC) realizes the payload connection without reproducing native seed expansion inside the application circuit. Prio3Histogram validity restricts the reconstructed contribution to a one-hot vector with a sentinel. On that domain, the linear index moment identifies the selected cell. Each role computes a masked receipt from the output it actually retained, and admission checks the receipts against the hidden-state proof. The binding relation is independent of histogram width, although native vector processing remains $O(D)$.

This bridge introduces receipts and retry feedback that neither component's isolated analysis covers. Authenticated source registration isolates the hidden input, while joint simulation couples receipts, errors, aborts and retries to the native transcript. Atomic admission rechecks the actual head, anchor and quota before commit. Each role applies authorized decisions once and durably reserves the complete eligible sum before sending its share. Figures 2 and 3 show these boundaries and their execution order. Integrity assumes correct services. Privacy separately permits one actively corrupted aggregator colluding with the Collector under declared metadata and equal complete-cohort aggregates.

The central contribution is the composition of an accepted-history contract, native-output binding, source and feedback protection, and reservation before first exposure. The contributions are as follows.

1) *Accepted-history contract.* A security contract connects hidden claims, actual native outputs, legal state advancement and complete membership at first exposure. Separating executions identify the missing semantic, history and exposure information.
2) *Compact native-output binding.* A masked-moment adaptor uses Prio3Histogram validity to connect hidden state proofs to retained outputs. Its binding relation is independent of histogram width and leaves native seed expansion outside the circuit.
3) *Privacy and recovery composition.* Source isolation and joint simulation account for receipt feedback. Under their respective models, the analysis establishes observer privacy and history closure across atomic admission, durable reservation and retries.
4) *Native realization and evidence.* A Janus implementation exercises semantic substitution, source-feedback and subset-exposure controls together with durable recovery. Materialized circuits and mechanism-level measurements characterize computational costs.

Section 2 defines the contract, Section 3 presents the construction and Sections 4 and 5 give its analysis and evaluation. Related work, limitations and artifact information follow.

2. Model and Security Contract

This section defines the deployment, accepted histories and release contract.

2.1. Deployment and Trust Model

Enrolled clients report under stable pseudonymous identities. An admission authority verifies hidden state proofs and maintains accepted history without receiving coordinate openings or role decryption keys. Two Aggregators prepare and retain Prio3 outputs. The Collector holds the aggregate-share HPKE key, and its Decoder verifies release records and reconstructs the heatmap. The authority-side Finalizer authenticates the frozen reference and its completion. Its certificate joins that reference to both roles' matching attestations rather than introducing another trust anchor. A durable admission proxy belongs to the same trust domain and delivers signed decisions. Each role durably reserves its own share before export.

Integrity and privacy use separate corruption models. For integrity, clients and transport are malicious, while admission, both Aggregators, Finalizer, Collector and Decoder execute the protocol. This model fits a managed domain whose operator enforces admission and release rules against deviating clients and unreliable transport. Serialized durable storage preserves committed history, retained outputs and reservations across recovery. Correct service execution does not give the authority coordinate openings. For privacy, one Aggregator may be actively corrupted with the Collector. The authority, honest clients, Finalizer and Decoder follow the protocol. The observer may also receive the correct authority's business records, receipts, diagnostics and backups. Sharing does not transfer its signing key or authorize malicious issuance. A corrupted role can fabricate shares or abort, so this model makes no output-integrity claim. Corrupt admission is outside both guarantees.

Privacy challenges use legal histories with common public structure μ, including identities, task and key, windows, step and anchor references, eligibility, contribution bits, packet rules and the delivery plan. Proofs, ciphertexts, commitments, receipts, errors, aborts and retries are simulated jointly. The authority has no aggregate interface. The aggregation observer has one visible cohort per task and key, with equal declared native-index sums on that complete cohort. Definition 6 specifies the views and the first-exposure interface. Activity, contribution bits and cadence remain public. The permitted exact aggregate can itself support inference from auxiliary knowledge. The claim bounds additional protocol leakage rather than inference from that output and μ. It does not establish privacy for multiple

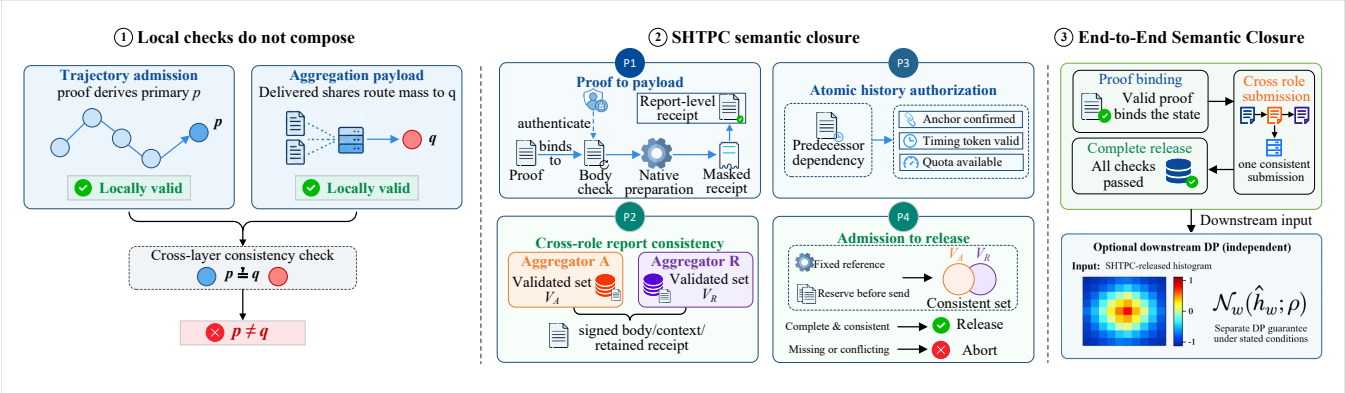


Figure 2. SHTPC connects report meaning, accepted history and complete release through four semantic boundaries.

outputs, including nonoverlapping windows, or hide network traffic beyond μ.

Correctly sampled setup with unknown trapdoors, reviewed fixed-width arithmetic, authenticated confidential registration and the declared durable execution boundaries are premises. Enrollment authenticates identity rather than physical location or Sybil resistance. Continuity uses authenticated slots rather than unverified elapsed time. No transparency mechanism detects authority equivocation in this model.

2.2. Reports, Histories and Cohorts

A sampling slot assigns a measurement to time, an accepted step counts a committed transition, and a release window groups eligible reports. Rejected attempts do not advance accepted history.

Each report is projected and quantized under fixed public parameters, yielding a hidden vector $\mathbf{x}_r$. A $W \times H$ grid with cell width s selects

$$p_r = \mathsf{Grid}(\mathbf{x}_r) = W \left\lfloor \frac{y_r}{s} \right\rfloor + \left\lfloor \frac{x_r}{s} \right\rfloor, \tag{1}$$

where $D = WH$. The evaluated profile uses $s = 100$ and $W = H = 100$. The replay projection, rounding and boundary rules appear in Appendix A.

Consider two drivers whose reports are registered for the same window. The candidate map P_w fixes their registered core records before preparation feedback, and $R_w = \mathsf{eligible}(P_w)$ selects the eligible cores. A committed record extends its core with authenticated receipts, decision and retained-output links. Projection π_{reg} removes those later fields. Admission records committed transitions in L_w, whose eligible projection is $\mathcal{A}_w = \mathsf{eligible}(L_w)$. Each role's map $M_{S,w}$ records the eligible decisions it has applied. If one delivery is delayed, the role's map is incomplete even though its local native batch is valid. Release compares that map with the fixed R_w, rather than using the local batch as the cohort.

Warmups enter the log but not the eligible maps. Eligible zero-bit records remain members despite adding no mass. Credential epochs and storage revisions do not replace accepted-step counters or sampling slots.

TABLE 1. NATIVE WINDOW LIFECYCLE AND SCOPE.

Stage	Contract
Register	All predeclared UID and contiguous step slots, including warmups and eligible zeros, need full pending bodies.
Lock	At the registration boundary, the complete graph starts at actual heads. Both roles install the lock.
Process	Current-head permits precede preparation and commit. Both roles apply decisions.
Expose	Complete records are reserved before the first share. Retries use the same snapshot.
Abort	Partial commits and quota persist. Later configured slots and new bodies start at actual heads.

For report r, the common admitted object is

$$A_r = (\mathsf{rid}_r, \eta_r, \mathsf{stmt}_r, \mathsf{pay}_r, c_{r-1}, c_r, w, b_r). \tag{2}$$

The context η_r binds identity, window and policy. The remaining fields identify the proof statement, bound channel packages, predecessor, successor and contribution bit. A complete signed body, two retained-output receipts and one commit decision authenticate this object in the native realization (Appendix B).

Strict release produces the complete eligible cohort or refuses. The native prototype uses a finite preconfigured schedule, not a roster selected from arrivals. Table 1 gives its window lifecycle.

After an abort leaves two owners at steps 2 and 1, their next configured slots start at 3 and 2 with new bodies referencing those heads. An incompatible schedule cannot lock. Configuration remains bound to the database. This integrity continuation does not extend the single-cohort privacy experiment.

2.3. Accepted-History Closure

Four obligations share A_r. P1 connects the proved claim to the consumed package by

$$\mathsf{Contrib}(\mathsf{pay}_{r,\mathrm{consumed}}) = b_r \mathbf{e}_{p_r}. \tag{3}$$

P2 requires both roles to consume channels of that same object. P3 gives each accepted object one atomic transition with its quota charge.

$$\mathrm{State}_{r-1} \to \mathrm{State}_r \tag{4}$$

P4 requires a successful release to contain exactly the eligible committed records and the complete locked reference.

$$\pi_{\mathrm{reg}}(\mathcal{A}_w) = R_w. \tag{5}$$

Together these obligations express accepted-history closure.

$$\mathrm{P1} \wedge \mathrm{P2} \wedge \mathrm{P3} \wedge \mathrm{P4} \;\Rightarrow\; \text{Accepted-History Closure}. \tag{6}$$

Definition 1 (Legal sequential history)**.** A sequence $H = (A_1, \ldots, A_n)$ with unique identifiers is legal if replay from enrolled genesis enables every transition. Each step has an authorized owner and context, names the current head and policy-selected accepted anchor, and satisfies the application relation on opened states. It atomically appends the successor, advances the head and accepted-step counter, and charges quota without changing other owners. Rejected and uncommitted attempts add no step.

Definition 2 (History refinement)**.** For a finite prefix, $\mathcal{L} \preceq H$ when committed admission operations bijectively map to a legal H, preserving every A_r field and the order of linearization points between invocation and response. Committed pending operations are completed at the prefix end, while uncommitted attempts are discarded. Retries of one durable decision represent one operation, and distinct committed effects remain distinct.

Definition 3 (Successful-trace closure)**.** Fix a transition policy, additive contribution map $r \mapsto m_r$ and eligibility rule. A successful release has closure if some legal H satisfies $\mathcal{L} \preceq H$. Let $E_w(H)$ be the set of records in the complete accepted history H that belong to window w and satisfy the eligibility rule. Their identifiers equal $\mathrm{dom}(\mathcal{A}_w)$, and each identifies one common admitted object across proof, packages, transition and certificate. Each role has $N_{S,w}(r) = \mathbf{1}[r \in \mathrm{dom}(\mathcal{A}_w)]$, and its reserved sum uses exactly those retained outputs. The pre-noise output is $\sum_{r \in \mathrm{dom}(\mathcal{A}_w)} m_r$. For heatmaps, $m_r = b_r \mathbf{e}_{\mathsf{Grid}(\mathbf{x}_r)}$. Eligibility is independent of bit value. These are policy-consistent admitted claims rather than verified physical locations.

Definition 4 (Exposure-prefix safety)**.** Every decryptable honest share comes from a prior durable reservation satisfying

$$\pi_{\mathrm{reg}}(L_{S,w}) = P_w, \qquad \pi_{\mathrm{reg}}(M_{S,w}) = R_w = \pi_{\mathrm{reg}}(\mathcal{A}_w),$$
$$\mathbf{a}_{S,w} = \sum_{r \in R_w} v_{S,r}. \tag{7}$$

Retries preserve its membership and plaintext. Successful-trace closure also requires correct final reconstruction.

Independent local checks do not supply these relations. For fixed context η, let $\mathcal{A}_p^\eta$ and $\mathcal{G}_p^\eta$ contain application and payload objects with primary p.

Proposition 1 (Role-separability observation)**.** *Independent acceptance is a Cartesian product. The same-primary relation $\bigcup_p \mathcal{A}_p^\eta \times \mathcal{G}_p^\eta$ is not a product when two primaries are feasible on both sides.*

The observation identifies the missing correlation rather than ruling out a shared tag or joint relation [15]. Appendix A gives the short proof.

Proposition 2 (History and release separation)**.** *A proof for predecessor c remains valid after a competing successor commits. Receivers seeing only R also cannot distinguish accepted set R from $R \cup \{r^\star\}$ when $r^\star$ is dropped on both channels. Checks on those identical views cannot accept the complete case and reject the incomplete one.*

Current-head evidence and an authenticated complete-set reference supply the missing history-dependent information without fixing a database or proof format. Appendix A formalizes the views.

3. SHTPC Design

This section describes how SHTPC realizes P1–P4 in a native Prio3 workflow.

3.1. Design Overview

Figure 3 follows one report from registration to exposure. The client proves Anchor Distance Window Continuity (ADWC) and grid membership over hidden states. Admission locks the registered graph before private preparation, checks both retained-output receipts, and commits against the current history. A durable proxy delivers that decision for exactly-once application. Each role reserves the complete eligible sum before sending its share. Preparation creates a retained output, while admission authorizes its use in aggregation.

3.2. Binding Hidden State to Native Outputs

P1 connects the certified cell to the output actually retained for aggregation. Generic vector binding proves that retained shares reconstruct the certified vector. SHTPC instead identifies that vector within the native one-hot domain, leaving seed expansion outside the application circuit. Table 2 compares these choices.

A direct binding proof could reproduce native sharding inside the application circuit. SHTPC instead reuses the domain established by native validity. For example, a three-cell one-hot vector $(0, 0, 1)$ has index-weighted sum $0 \cdot 0 + 1 \cdot 0 + 2 \cdot 1 = 2$. Thus its moment identifies the selected cell. By linearity, the moments of the roles' retained shares combine to identify the reconstructed one-hot vector. Arbitrary vectors can share a moment.

The public transition statement contains the relation version, predecessor, successor and anchor commitments and aliases, mask, ticket and receipt commitments, key tag, masked index, bit, report identifier and context. Its witness

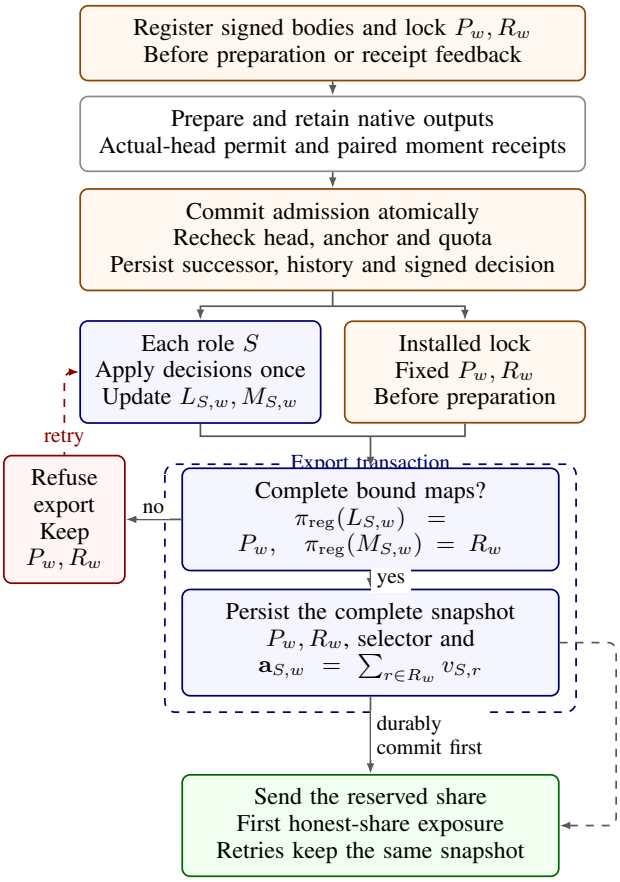


Figure 3. SHTPC reserves the complete eligible history before share exposure.

TABLE 2. BINDING DESIGNS ESTABLISH DIFFERENT RELATIONS UNDER THE SAME RELEASE CONTRACT.

Approach	Certified connection	Evidence or remaining work
Association only	Common report or committed value	No retained-output equality
Direct seed binding	Seed expansion and selected vector	Measured R1CS size
Generic vector binding	Retained shares equal certified vector	Measured transition cost (this work)
Masked moment	Certified index equals retained index	Measured R1CS size and native controls

opens those commitments and gives the coordinates, grid cells, remainders, masks and carry. The relation binds the payload witness's cell and remainder to the successor and enforces $p = \mathsf{Grid}(\mathbf{x}_r)$, $\|\mathbf{x}_r - \mathbf{x}_{\mathrm{prev}}\|^2 \le \tau_u^2 \Delta t_t^2$ and $\|\mathbf{x}_r - \mathbf{a}_t\|^2 \le \min\{\tau_u^2(\Delta T_t^{(K)})^2, c_{\mathrm{policy}}^2\}$. Here $\mathbf{a}_t$ is the policy-selected accepted anchor, τ_u is the speed tier and c_{policy} is the anchor cap. Admission checks that the referenced predecessor and anchor are the actual accepted ones. The same grid and distance constraints apply when $b = 0$. Appendix A gives the parameters, and the artifact's fixed transition circuit gives the ordered public-signal interface (Section 8). In native field $\mathbb{F}_q$, use $D < q$ and $D+1$ buckets with sentinel D. A state-derived primary $p < D$ and public bit b authorize $x = bp + (1-b)D$. Native validity gives $v_A + v_R = \mathbf{e}_y$ for some $y \in [0, D]$.

The client samples private uniform masks r_A, r_R, commits to their openings and encrypts one for each role. The proof binds the primary, bit, masks and completed-package context to

$$T = x + r_A + r_R \pmod q,$$
$$m_S = \sum_{j=0}^{D} j v_S[j] + r_S. \tag{8}$$

Each correct role computes m_S from its retained output. Admission checks report, context and statement identities and accepts the paired receipts only if $m_A + m_R = T$. Lemma 1 shows that acceptance forces $y = x$, giving $b\mathbf{e}_p$ after sentinel removal. Canonical scalars enforce $x + r_A + r_R = T + \chi q$ with $\chi \in \{0, 1, 2\}$, so the different proof and native fields cannot wrap the equality (Appendix B). Coordinate openings remain client witnesses. Reused state points preserve their birth domains and exact accepted references. Receivers recompute the authenticated package context from the actual bytes. The authority checks the masked equality without receiving either coordinate opening or a native role key.

The binding relation is independent of D, while native vector processing remains $O(D)$. The adaptor separates preparation from accumulation. Each signed receipt binds the body, configuration, report, native job, terminal wire step and retained-session identifier; exact retries preserve output and receipt. The opaque session tag identifies the local output, while the body and report context pair the roles around the same admitted object.

3.3. Protecting the Feedback Interface

Encryption does not authenticate an input's owner. An attacker can transplant an honest ciphertext and substitute a known mask, causing an honest receipt to reveal the cell before admission rejects the report. This feedback arises from the new bridge even when the proof and native protocol are secure in isolation.

SHTPC allocates an owner identifier to one complete signed body and registers it before private processing. Each ingress authenticates the actual metadata, ciphertexts, sidecars, installed lock and permit against that body. The implementation stores the full body, and ingress checks its actual consumed bytes. Both retained sessions and receipts therefore name the same admitted object, realizing P2.

Recovery retains the same source and native state. The Helper journals its first protected input and response before sending. Identical retries replay those bytes, while changed inputs or new jobs cannot reopen that branch. The Leader's lease and transaction fence permit at most one committed transition. Uncommitted computation can receive new peer feedback, which the joint privacy analysis includes explicitly.

3.4. Atomic History Authorization

P3 makes proof validity current at commit. Admission verifies the proof before the transaction, then atomically rechecks the actual head, anchor and quota and stores the successor, history and signed decision. Both retained-output checks precede this linearization point. Preparation, admission and role application use separate databases connected by that decision and its output links.

For stable identity u, task and window w, durable quota $q_{u,w} \in \{0,1\}$ is shared across credential epochs. With accepted round t and $N_{\mathrm{warm}} = \max(K,2)$,

$$\begin{aligned} \gamma_t &= b_t \mathbf{1}[t > N_{\mathrm{warm}}], \\ q_{u,w} + \gamma_t \le 1, &\qquad q_{u,w} \leftarrow q_{u,w} + \gamma_t. \end{aligned} \tag{9}$$

Commit induction gives

$$q_{u,w} = \sum_{\substack{r \in \mathcal{A}_w \\ uid(r)=u}} b_r \le 1. \tag{10}$$

Warmups remain in the transition log without charge. Eligible zero-bit updates remain members and can advance state after quota is spent.

History is keyed by stable $(\mathsf{task}, \mathsf{uid})$, with credential, revision, window, receipt and budget checks inside the transaction. The proxy queues exact decisions durably before delivery. Role databases apply each decision once, including out-of-order delivery. Retries return that decision, and credential rotation preserves history and spent quota.

3.5. Complete Exposure and Recovery

P4 fixes membership before preparation feedback can influence it. Admission locks the verified candidate graph and its eligible projection at the public registration boundary. The graph includes warmups and eligible zeros together with their predecessor and anchor references. Preparation errors and receipts cannot select a smaller cohort or replace a locked body. Export by role S requires

$$\pi_{\mathrm{reg}}(L_{S,w}) = P_w, \qquad \pi_{\mathrm{reg}}(M_{S,w}) = R_w. \tag{11}$$

These equalities compare the frozen registration cores, including context and body. The role separately checks each actual retained-output link against its authenticated receipt and commit decision. Matching histogram values do not replace membership checks.

In one export transaction, the role checks these maps and reserves the selector, record map, retained-output links and plaintext sum. Only then can it send the aggregate share. Warmup completion is checked through $L_{S,w}$, and eligible zeros remain in $M_{S,w}$. Responses and re-encrypted retries read the same durable snapshot, so recovery cannot change its plaintext or membership. The Finalizer authenticates the frozen reference and both roles' matching attestations. The Decoder checks that certificate and the reserved records before reconstruction. Numerical equality alone would not establish this membership because an omitted eligible zero has no displayed mass.

TABLE 3. SECURITY PROPERTIES AND OBSERVER VIEWS RELY ON DISTINCT MECHANISMS AND PREMISES.

Property or observer	Main mechanism	Premise or permitted output
Legal hidden witness	Groth16 extraction, Pedersen binding	Unknown trapdoors and typed generic-group extraction
Proved cell is counted	One-hot validity, masked moments	Correct retained outputs and $D < q$
Legal history step	Authentication, commit induction	Atomic head and quota update with durability
Complete first exposure	Lock, apply and reserve	Lock before feedback and persist before send
Authority privacy	Blinding, ZK, HPKE and receipt simulation	Common metadata and no aggregate interface
Role and Collector privacy	Full-verifier and joint-view simulation	Static corruption, one cohort and equal full sums
Combined observer	Joint business-view simulation	Same complete output and no authority signing key

After locking, an inconsistent or permanently absent member can hold or abort the cohort without removal. Committed decisions survive delivery failure and are retried without another client upload. Collection failure preserves the first exposure and reserved snapshot. Pre-exposure abort retains partial commits, heads and quota for continuation as in Table 1.

For a finite fully committed lock, eventual service and channel recovery with fair retries and no deadline stop gives complete release. Each successful apply removes one of at most $2|P_w|$ pending role-record pairs. Permanent client refusal and incomplete pre-lock registration fall outside these progress conditions. Admission caps bound accepted work and exact retries consume no new slot. The native profile uses $B_u = 16$ per owner and window.

The strict profile releases an exact heatmap rather than a differentially private output. Its single-cohort privacy scope is in Section 2.1.

4. Security Analysis

This section analyzes history closure and privacy under their respective trust conditions.

4.1. Security Guarantees

Table 3 separates the premises for history closure from those for observer privacy.

Definition 5 (Closure game)**.** The challenger initializes honest service keys, enrolled identities and genesis state. A bounded PPT adversary controls malicious clients and delivery. It submits packages, observes responses, replays,

replaces, reorders or suppresses messages, and invokes admission and window closure. Honest signing keys and committed service state remain outside its control. The challenger records the locked P_w, R_w, invocation and response events, committed objects and receiver application counts. The adversary wins if the Decoder accepts a release for which no legal linearization exists, or any membership, application-count, common-object or reconstruction condition of Definition 3 fails, or if a native role exports a share that violates Eq. (7). Refusal without an unsafe export is not a win. The game fixes the strict profile. Write $\mathrm{Adv}_{\mathrm{closure}}(\mathcal{B}) = \Pr[\mathsf{Win}]$. Serializable state and correct service computation are premises of this game, not events assigned a cryptographic failure probability.

Lemma 1 (Native moment fidelity)**.** *Assume sound relation certification, commitment binding, authenticated common body identity, correct receivers and native validity. Let the certified canonical coordinate be $\mathbf{x}_r$, with $p = \mathsf{Grid}(\mathbf{x}_r) < D < q$ and $b \in \{0,1\}$. Let $x = bp + (1-b)D$, with canonical $r_A, r_R, T \in [0, q-1]$, and $F(v) = \sum_{j=0}^{D} j v_j$. Suppose the retained outputs satisfy $v_A + v_R = \mathbf{e}_y$ and the receipts satisfy $m_S = F(v_S) + r_S$. If certification fixes $T = x + r_A + r_R \pmod q$ and admission checks $m_A + m_R = T$, the reconstructed contribution after sentinel removal is $b\mathbf{e}_p$.*

Proof. Opening checks fix the certified masks. Linearity gives $y = F(v_A + v_R) = T - r_A - r_R = x \pmod q$. Since $x, y \in [0, D]$ and $D < q$, this is integer equality. Removing sentinel D gives $b\mathbf{e}_p$. Theorem 1 accounts for the cryptographic and native-validity failure events, while correct receiver execution remains a premise. □

Proposition 3 (First-exposure separation)**.** *Under the share-observation interface of Section 2.1, suppose two metadata-compatible admissible histories have the same aggregate on a full accepted set $\mathcal{C}$, but different aggregates on a permitted native batch $R \subsetneq \mathcal{C}$. Native validity, atomic admission and exact-set checking at final publication do not imply exposure-prefix safety or privacy relative to the full aggregate. This holds even for a wrapper that successfully publishes $\mathcal{C}$ on complete delivery.*

The separating execution delivers complementary shares of R to the corrupted role and Collector, then refuses publication because $R \neq \mathcal{C}$. Appendix A constructs the two legal histories. The exposure distinguishes the histories despite the later publication refusal.

Related-setup extraction. The typed generic bilinear model permits source-group linear operations and pairing, without a target-to-source map. The analyzed public basis includes raw and prepared phase-one points, initial and final proving keys, verification keys and contribution proofs for current genesis, current transition and historical genesis. These share τ, α, β, use $\gamma = 1$, and have three independently sampled phase-two deltas Δ_j. Other imports are independent uniform exponent points or known-scalar combinations of existing handles. Write $q_{j,i} = \beta u_{j,i}(\tau) + \alpha v_{j,i}(\tau) + w_{j,i}(\tau)$. Public IC contains $[q_{j,i}]_1$, while the final key contains $[q_{j,i}/\Delta_j]_1$ for private wires. The enumerated basis has no such inverse point for a public wire, which would enable statement transport. A later eight-bin experimental key used the same phase one. If that key or other setup-dependent material outside this basis is available to the adversary, the three-key bound needs a new source ledger even if the runtime rejects it.

External real and simulated source statements are registered before fresh coins, with identity given by key namespace, relation and ordered signals. Honest sources reuse their registered semantics, known-witness sources retain their witness, and newly extracted witnesses are cached. Foreign sources remain budgeted auxiliary inputs and cannot become current-namespace source exceptions. Malicious owners receive no witness-free proof service.

Lemma 2 (Related-setup extraction)**.** *Under these rules, an accepted proof yields a witness or names a registered source with identical key, relation and ordered signals, except with probability at most g. Appendix B gives the finite source and generic-operation budget for g. The complete derivation, including the source inventory and both extraction directions, is in artifact proof document G4b 02, Sections 2–7 (Section 8).*

Proof sketch. In exponent notation, log each source proof as $A_i = \mu_i$, $B_i = \nu_i$ and $C_i = (\mu_i\nu_i - \alpha\beta - Q_j(\phi_i))/\Delta_j$, where $Q_j(\phi) = \sum_{i\,\mathrm{public}} \phi_i q_{j,i}$ includes the constant wire. Comparing coefficients rules out a source C_i in candidate A. Matching source μ_i, ν_i then forces the same delta and statement. Without source terms, the two possible coefficient directions both extract the target private assignment. The appended six genesis and 26 transition public A-only unit rows separate public from private u-polynomials. At a unit-row root the residual vanishes, excluding the extra τ^{n-1} direction of the transition H-basis and making the extracted residual divisible by the target vanishing polynomial. Extraction is online without rewinding. □

Atomic commits witness Definition 2. Durable receiver indexes enforce the application counts in Definition 3. Appendix A connects both invariants to export.

Theorem 1 (Accepted-history composition at first exposure)**.** *Under the integrity model of Section 2.1, let B_C bound whole-execution proof extraction, commitment binding, authentication, digest and origin failures, and let v_N bound native validity failure. Appendix B derives these terms from the specified primitives and finite query budgets. Then*

$$\mathrm{Adv}_{\mathrm{closure}} \leq \min\{1, B_C + v_N\}. \tag{12}$$

Outside those events, every first export and retry satisfies Eq. (7) with one immutable snapshot. The accepted log refines a legal ADWC history under Definitions 1 and 2. Every successful strict release satisfies Definition 3 and reconstructs

$$\hat{\mathbf{h}}_w = \sum_{r \in R_w} b_r \mathbf{e}_{\mathsf{Grid}(\mathbf{x}_r)}. \tag{13}$$

Each record has one common admission object across its proof, consumed packages, committed transition and release certificate. These invariants persist through recovery and collection failure.

Proof sketch. Under the source rules of Lemma 2, honest same-source reuse retains its registered semantics, malicious sources use known or extracted witnesses, and foreign namespaces or contexts are rejected. Commitment binding fixes the predecessor and anchor. Native validity and Lemma 1 bind its retained output to the proved cell. Induction over registration, lock, permit, commit and apply preserves the fixed candidate maps and a legal accepted history. The export transaction checks full completion and fixes the share before sending. Recovery preserves that transaction's snapshot. The roles apply the durable certificate idempotently without a cross-database atomic commit. After a correct freeze, complete membership is stable under monotone application, so an earlier incomplete snapshot can refuse but cannot authorize a smaller set. Consequently every exposure prefix satisfies Eq. (7), and reconstruction gives Eq. (13). Appendix A gives the event cases. □

Equation (13) also supplies the admitted-history numeric input to any downstream mechanism, without changing that mechanism's own privacy or output policy.

In Eq. (12), g bounds failure of the related-setup witness-or-source classification of Lemma 2 for Groth16 [16]. The term c_B bounds conflicting Pedersen openings, and a_C bounds fresh signature forgeries on keys the adversary does not hold. The term ϵ_H bounds digest collisions between distinct canonical records. The origin term bounds guessing an honest identifier before publication. All budgets count failed attempts, offline work and retries. The remaining term v_N bounds acceptance of an invalid native input over the seven-purpose stream query budget. The public verification key tag is a 96-bit prefix, so without a further assumption $\kappa_{\mathrm{tag}} \le \min\{1, 2^{96}G_5/2^{128}\} = \min\{1, G_5/2^{32}\}$ for guessing the 128-bit key, not colliding tags. The alternative branch needs explicit SHA256 one-wayness on uniform 128-bit inputs and does not follow from collision resistance (Appendix B). This tag term is a conservative native-validity bound, not a measured attack probability or a privacy term. The remaining primitive advantages stay symbolic because the analysis does not assign them a common numerical security level.

Definition 6 (Single-cohort privacy experiment)**.** Fix a finite public resource budget before setup. Sample a correct setup and fix the task, profile, keys and static corruption set before any honest measurement. The budget covers honest and adversarial work, including warmups, zero-bit reports, failed and unprepared reports, verification, oracle calls, bytes and retries. Execution stops at the budget. The environment supplies two complete legal histories H_0, H_1 for honest sources with the same public skeleton μ. It fixes each source's challenge inputs before its first report, while the adversary can adaptively request reports from those histories. Honest reports reach admission over an independent authenticated confidential channel according to the input-ready plan. Local witness and proof generation time is outside the online view.

For $X \in \{A, P, P{+}A\}$, the challenger selects a uniform hidden bit d, runs the protocol on H_d, and interacts with a bounded PPT adversary $\mathcal{B}$ with the following view. In A, the correct authority's view includes its business records, receipts, diagnostics, database and backups, and its ordinary authorization secrets. It has no role decryption keys or aggregate output. The observer may compute offline with its keys but cannot use them to issue extra authority signatures, change the configuration or bypass the state machine. In P, one actively corrupted Aggregator and the Collector share their state, decrypted inputs, peer messages and actual available Collector results. The corrupted role can fabricate messages and shares or abort. In $P{+}A$, that observer also sees the correct authority's business view, without its authorization secret or malicious issuance. The adversary can control corrupted owners and schedule, delay, replay or drop messages within the model. Each view includes its visible proofs, ciphertexts, commitments, responses, refusals, retries, aborts and modeled online times. These outcomes are not assumed to be part of μ.

For each record, let $y_i^{(d)} = b_i p_i^{(d)} + (1 - b_i)D$ denote its declared native index, where $p_i^{(d)}$ is its declared primary cell. For each allowed complete cohort R_w, the ideal interface returns the declared native-index sum $\mathcal{O}_d(R_w) = \sum_{i \in R_w} \mathbf{e}_{y_i^{(d)}}$, including sentinel D. Write $\mathcal{O}(R_w)$ when the world is clear. This sum is not necessarily the actual Collector result when a role or source deviates. The A simulator receives no aggregate query and its two histories need not have equal sums. The P and $P{+}A$ histories have equal declared sums for every allowed complete cohort. Their simulators may query that full cohort once at the boundary where the first honest share becomes decryptable, even if collection later fails. Before that boundary they receive no output, and retries reuse the reserved plaintext without another query. No subset query is permitted. The game covers one visible selector for the fixed task and key.

The adversary outputs a bit in Exp_X^d. Its distinguishing advantage is $\mathrm{Adv}_X(\mathcal{B}) = |\Pr[\mathsf{Exp}_X^1(\mathcal{B}) = 1] - \Pr[\mathsf{Exp}_X^0(\mathcal{B}) = 1]|$.

Theorem 2 (Joint privacy of admission and aggregation views)**.** *Fix the privacy experiment of Definition 6 with static corruption, the strict profile, and the source and programmable-oracle models in Appendix B. The authority view is simulatable without honest coordinates or an aggregate. The aggregation view, including any shared authority business view, is simulatable from the observer's inputs, μ and the permitted complete-cohort sum. The distinguishing advantage of each observer X is at most $\mathrm{Adv}_X \le \min\{1, 2e_X\}$, where e_X composes the primitive bounds above, the hidden HPKE encapsulations and the native stream simulation. Theorem 4 gives the concrete form and a hybrid outline.*

The masked receipts provide the key link. Let c be the corrupted role and h the honest role, and let v_c^{orig} be the

native output determined by the corrupt role's original input. For an honest source,

$$m_h = T - F(v_c^{\text{orig}}) - r_c. \tag{14}$$

The simulator can choose uniform T and compute this receipt before knowing the cell. An arbitrary corrupt receipt z passes the moment check exactly when $z = F(v_c^{\text{orig}}) + r_c$, which is already computable from the corrupt input. For the authority alone, uniform T, U give $(m_A, m_R) = (U, T - U)$ without any output query. These identities join the receipt simulation to the native preparation transcript and the same causal history state.

The simulator first classifies each request by its registered body and exact statement, including the key namespace and ordered public signals. Honest same-source reuse preserves registered business semantics and its native handle. Malicious sources use known or extracted witnesses. Foreign namespaces or contexts are rejected. A changed honest body under the same statement requires a digest collision or an unauthorized signature. For a fresh statement, extraction supplies the committed openings, not the ciphertext plaintext. A ciphertext distinct from the protected challenge tuple is handled by legitimate decryption and native execution. This routing permits malicious sources without providing an oracle for an honest source's challenge.

For an honest source, the coupled receipt in Eq. (14) uses the same native share as the preparation transcript. For a malicious source, let y_j be its extracted or known claimed index and $w_{h,j}$ the actual successful output of the honest role. The simulator does not assume that these outputs satisfy the honest-source identity. When reservation permits first exposure, the single complete-cohort query $\mathcal{O}(R_w)$ gives

$$\begin{aligned} V_{h,R_w} = \mathcal{O}(R_w) &- \sum_{j \in R_w \setminus H} \mathbf{e}_{y_j} \\ &- \sum_{i \in R_w \cap H} v_{c,i}^{\text{orig}} + \sum_{j \in R_w \setminus H} w_{h,j}. \end{aligned} \tag{15}$$

Here H is the set of honest native sources. The oracle returns the sum of declared indices, including the sentinel, rather than the Collector's actual output. Subtracting the known or extracted malicious-source declarations leaves the honest-source sum. For each honest source, $v_{h,i} = \mathbf{e}_{y_i} - v_{c,i}^{\text{orig}}$. Subtracting the original corrupt-role shares therefore gives their honest-role aggregate. Adding the actual honest-role outputs $w_{h,j}$ for malicious sources completes the joint view without assuming that those sources follow their declarations. This uses no subset query. Retries reuse the reserved plaintext. The authority-only simulator uses the zero-query receipt law above.

Appendix B gives the concrete Histogram kernel and reduction budgets for this joint simulation. The reviewed G5 receipt core uses two fixed scalar multiplications, one point addition, canonical range checks and constant-time four-element equality, with no secret-dependent branch or address in payload equality. Correct setup and durable state remain premises of Theorems 1 and 4.

5. Evaluation

This section evaluates application impact, security interfaces and costs.

5.1. Setup and Comparison Contract

Research questions. The evaluation asks how report substitution and omission affect heatmaps and release availability, whether the tested interfaces enforce binding and complete exposure, and what costs the construction incurs.

Datasets and workloads. Controlled attack replay uses 100 selected legal histories each from T-Drive[1] and Porto[2], aligned by relative step with three fixed actor permutations. A separate input screen uses T-Drive, Porto, GeoLife [17] and Rome [18]. It pools three frozen dates per corpus without prefiltering actors by legality. GeoLife and Rome dates were selected by observed actor coverage. The unit is an observed actor-day, so actors recurring on different dates are not independent users. The screen uses 300-second slots and a 100-m grid spanning 10 km, with dataset-specific centers and trajectory boundaries in Appendix A.

Implementation and setup. SHTPC uses Janus [19] with two-role Prio3Histogram under the implementation's DAP draft [5, 20], 10,000 spatial cells and a sentinel. The server-window mechanism ablation uses 16 logical CPUs on an Intel Xeon Gold 5218R at 2.10 GHz. The binding components and Full/T lifecycle pair use an Intel Xeon Gold 6148 at 2.40 GHz, with Node 22.22.1 and snarkjs 0.7.4 for the component measurements. The Full and trusted-generation pair runs sequentially on the same host and receiver on CPUs 0–15, with 100 identities, dimension 10,001, 600 priming and 100 eligible reports, no zero reports, at most four bodies per public drive, 16 workers and pipeline limit 64. Both use deterministic synthetic geometry and the same slot structure, with fresh identities and keys per run. Each arm has one observation.

Baselines and controls. Binding comparisons use direct seed expansion and vector commitment. The trusted-generation control T holds plaintext coordinates, state and randomness to generate reports, whereas Full clients generate transition proofs and moment sidecars. The receiver selects Full or T using authority-signed configuration. T verifies trusted receipts bound to the statement, report and context, while Full retains proof and sidecar checks. N8 controls exercise both paths and reject cross-mode and tampered inputs. The Prio3 reference uses a different native-component endpoint. SCIF and PEAR controls use different aggregation interfaces. Offline replay compares independent composition with a strict-contract oracle that assumes detection.

Metrics. Replay reports L1 error normalized by the original cohort and target-cell counts. Native tests report acceptance, release and recovery. Component measurements

1. https://www.microsoft.com/en-us/research/publication/t-drive-trajectory-data-sample/
2. https://archive.ics.uci.edu/dataset/339

TABLE 4. OFFLINE TARGET-CELL COUNTS UNDER 20% HOTSPOT SUBSTITUTION ACROSS THREE FIXED PERMUTATIONS.

Dataset	Clean	After substitution
T-Drive	3	22–23
Porto	2	21–22

use a fixed fixture per relation on one designated logical CPU, fresh Node processes, one excluded warmup per relation and five interleaved pairs. System costs distinguish the mechanism-ablation server-only window from the lifecycle window from client generation through collection and the full session.

5.2. Real-Trajectory Impact and Availability

Substituting 20 of 100 reports in offline independent composition produces 0.38–0.40 cohort-normalized L1 error. When substitutions target the clean modal cell, its count rises from 3 to 22–23 for T-Drive and from 2 to 21–22 for Porto (Table 4). The strict-contract oracle instead withholds every tested window with nonzero substitution or omission. This trades release availability for integrity, and an aborted window has no defined output error.

Janus runs on the first eight selected pressure histories from T-Drive and Porto release exact clean outputs and refuse both substituted reports in each 25% attack run before commit, with no incomplete-set export. There is one clean and one attack run per dataset. These points exercise native interfaces but do not turn the offline oracle sweep into native measurements.

Figure 4 separates spatial coverage, consecutive in-grid observations and policy legality using mutually exclusive outcomes. T-Drive is affected by continuity and the anchor, Porto principally by fragmented trips and continuity, and GeoLife by coverage of the fixed Beijing grid. The Rome cache fits this configuration well. The pooled legal counts are 12,162/29,658, 116/1,200, 9/85 and 836/871 observed actor-days, respectively. Different dates, centers and trajectory boundaries make these descriptive input measures rather than a ranking. Rome's frame contains observed actors in the cleaned city-point cache.

Individual legality and a complete simultaneous input are distinct. Of 15 predetermined Rome rosters per size, 11 at N=8 and two at N=32 have individually legal histories for every member, but none of these has a common start, giving 0/11 and 0/2. At N=100 no roster is individually all-legal and the conditional result is undefined. This eight-slot offline screen does not measure native release or exhaust possible schedules and accepted histories.

5.3. Compact Semantic Binding

The masked-moment check rejects a native-valid report for cell 42 when the hidden proof authorizes cell 2312. The authority records no commit, head advance or quota consumption. This exercises Lemma 1 on native-valid inputs.

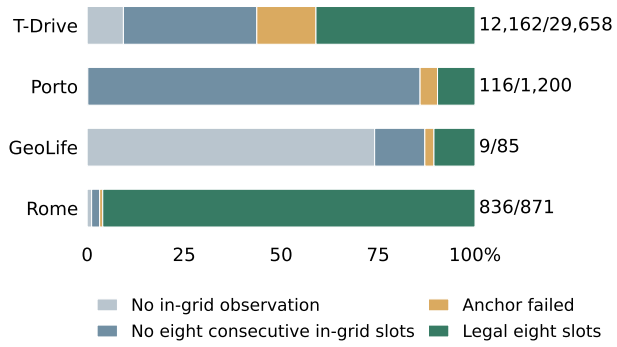


Figure 4. Offline input-screen outcomes across three observed actor-day frames per corpus.

TABLE 5. FIXED-FIXTURE TRANSITION COMPONENT COSTS ACROSS FIVE INTERLEAVED PAIRS.

Metric	Masked moment	Vector commitment
fullProve median (s)	3.4805	311.8310
fullProve range (s)	3.4674–3.4820	311.5520–312.4485
Prove CPU median (s)	3.4748	311.3832
Process peak RSS median (GiB)	0.2977	13.8623
Internal verify median (s)	0.03230	0.04054

At 10,001 bins, the standalone masked-moment module has 5,779 constraints versus 79,002,390 for direct seed binding, a 13,670.60× ratio (Figure 5). The complete transition measurements in Table 5 concern different relations and fixed fixtures, rather than the same business input. Setup verification preceded measurement. The five paired fullProve time ratios have median 89.73, quantifying component costs for these implementations. fullProve includes witness generation. Native moment evaluation scans the vector in $O(D)$, and the compact relation relies on native one-hot validity.

5.4. Feedback and First Exposure

Origin checks protect an earlier boundary. In the feedback control, disabling body-owner signature and authority-certificate checks allows copied honest reports with known masks to produce receipts that reveal index 4237. Later admission refusal does not erase this observation. The complete guard refuses the same input before decryption and receipt feedback. Normal reports and exact retries continue through the protected path. The result locates origin authentication before receipt feedback.

Two legal histories produce the same complete eight-owner histogram but produce different seven-owner subset outputs (Figure 6). The native subset control exposes a difference of two counts in unnormalized L1, the sum of absolute bucket-count differences. SHTPC refuses the incomplete set before an honest share is exposed, illustrating the first-exposure boundary in Proposition 3.

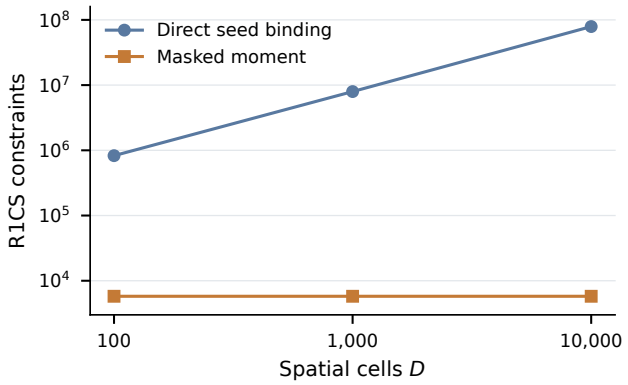


Figure 5. Standalone binding circuit R1CS sizes across histogram widths.

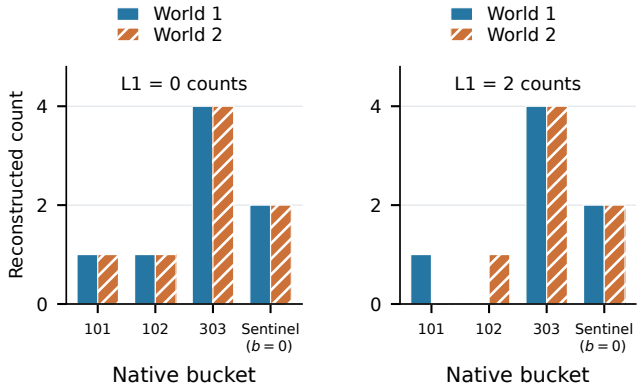


Figure 6. Native outputs for (a) the complete eight-member cohort and (b) its seven-member subset.

The external controls use different aggregation interfaces. A SCIF acceptance bridge detects omitted positives, eligible zeros and equal-count substitutions only after trusted output reconstruction, because that output carries no external member evidence [11]. A PEAR role run reconstructs an exact eight-report, 10,001-bin output and rejects invalid inputs without supplying SHTPC's current predecessor or accepted-set reference [12]. These controls complement rather than replace the native first-exposure check.

5.5. History Recovery

In the 100-user campaign, redelivery after one or ten withheld accepted decisions restores all 100 eligible members and the exact 10,001-coordinate output. The campaign fixes 700 transitions. Affected services restart on their retained databases while the key-holding controller remains live. Table 6 summarizes the observed obligations.

A two-owner native trace combines sequential competing registration, retries and selective delivery. The conflicting successor is refused, and redelivery restores the frozen set and exact output. The run passes seven driver and 469 independent semantic checks. Separate recovery tests

TABLE 6. NATIVE VALIDATION OF SECURITY OBLIGATIONS.

Obligation	Security boundary	Observed outcome
P1	Output binding	2312/42 mismatch rejected, no commit
P2	Feedback isolation	Guard-off revealed 4237, guard-on blocked
P3	Atomic history	Conflicting successor rejected, retry idempotent
P4	Complete exposure	Missing 1 or 10 blocked, 100-member output restored
P4	Subset exposure	Distinguishable seven-of-eight subset blocked

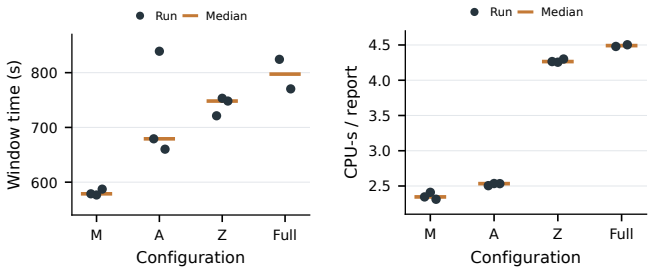


Figure 7. Mechanism costs for 100 eligible reports, with (a) server-window wall time and (b) server CPU per report.

cover concurrent preparation, changed peer bytes, arranged commit-first and freeze-first SQL orders, and failure paths that retain accepted state.

TLA+/TLC 2.19 checks 41 configurations of ClosureV5: fixtures have one to three owners and reports, two roles, and one or two windows. The artifact's `current/model/cases.json` fixes each case's bounds; its specification is `current/model/ClosureV5.tla`. Six safety and one fair-progress configurations pass; guard removals expose invariant violations, and reachability and temporal controls exercise witnesses and fairness assumptions. These bounded results and the synthetic-receipt correspondence tests complement, but do not replace, the protocol-level argument.

5.6. Cost of the Enforced Contract

Figure 7 reports four mechanism configurations for 100 eligible reports. The two Full server-window observations are 770.39 s and 824.36 s. Shared registration, lock, exposure and collection mean the configurations do not isolate additive increments or P4. M/A/Z have three runs and Full has two, with median lines in the figure. The separate Prio3 native-component reference has a 10.59 s median at a different endpoint.

Prio3 supplies native validity and aggregation. M adds masked-moment consistency, A adds permit and receipt authorization, quota and durable delivery, and Z proves the hidden transition. Full combines these mechanisms with atomic admission and complete first-exposure reservation. In

TABLE 7. SELECTED PHASE AND SESSION COSTS FOR THE FULL AND TRUSTED-GENERATION PAIR.

Phase	Full (s)	Trusted generation (s)
Setup and genesis	197.225	201.259
600 history priming	3322.162	2309.288
Eligible window (100 reports)	529.450	378.330
Total session	4342.151	3171.444

a separate server-only diagnostic, admission and verification account for the largest measured CPU category.

Full lifecycle and trusted generation.. Table 7 reports selected phase and session costs for the Full/T pair. Full takes 72.37 minutes for the session and 8.82 minutes for the eligible window, compared with 52.86 and 6.31 minutes under T. These single observations are lower under T by 26.96% and 28.54%, respectively, under a different trust boundary. Full does not disclose coordinates to the trusted generator. T omits transition proofs and moments but retains 100 genesis proofs. The window includes collection, which takes 57.062 s for Full and 65.326 s for T. The listed phases do not exhaust the session.

6. Related Work

Table 8 compares the guarantees and integration obligations most relevant to accepted-history closure.

Validation and inclusion. Prio+, VDAFs and DAP separate private input validation from aggregation and collection [4, 5, 27, 28]. Poplar, Mastic and Whisper extend private analytics to heavy hitters, weighted metrics and streams [29–31]. ELSA and RoFL address malicious contributions, while fully linear PCPs enable proofs on shared data [9, 10, 32]. VerSA and VeriFL study aggregate verification, with later analyses and repairs [13, 33–35]. Input validity constrains the reconstructed value, whereas inclusion constrains which reports participate. SCIF supplies honest-input inclusion under its server-corruption threshold. PEAR can express an application predicate, but using that predicate with evolving state also requires an authenticated current predecessor and atomic advancement. SHTPC connects these interfaces to an external accepted history before the first honest share becomes decryptable. In particular, matching counts or histogram values do not certify inclusion of an eligible zero-bit record.

State and authorization. Proof-carrying data and recursive composition propagate evidence of computation history [21, 36]. Privacy Pass, DAP report authentication, zkcreds and ZEBRA support authorization or private credential claims [37–40]. VPriv and PrETP connect private locations to charges [2, 3]. PrivStats combines anonymous reporting with per-point and total upload quotas and sample-range proofs. Its accountability goal is therefore closer to this work than origin authentication alone. SHTPC additionally authorizes a successor against the actual per-identity head and binds its retained contribution to complete first exposure. A transition proof can remain valid after another successor commits, so checking that proof and updating the head are distinct operations. SecAgg+ and LightSecAgg address dropout recovery [41, 42]; SHTPC instead preserves its prelocked membership through durable redelivery.

Output privacy. RAPPOR randomizes locally, while Prochlo separates encoding, shuffling and analysis [26, 43]. Shuffle-model analyses quantify amplification from anonymous randomized reports [44–46]. Differential privacy protects released statistics [47, 48], while geo-indistinguishability and temporal models address location disclosure and correlation [49–51]. Sparse heatmaps combine secure aggregation with distributed differential privacy, separating private computation from information in the released statistic. These goals complement SHTPC's guarantee that the statistic contains the admitted claims. Applying an output mechanism requires its own sensitivity, release policy and composition analysis, including which party can first decrypt an exact result. Its single-cohort simulation bounds additional protocol information under the declared output and metadata, rather than inference from successive released heatmaps.

TABLE 8. RELATED GUARANTEES AND THEIR CONNECTION TO ACCEPTED-HISTORY CLOSURE. THE LAST COLUMN IDENTIFIES INTEGRATION OBLIGATIONS, NOT ATTACKS OR MISSING GUARANTEES UNDER ANOTHER PROTOCOL'S MODEL.

Representative work	Established focus	Connection to this contract
Prio, ACORN, EIFFeL [6–8]	Validity of private inputs	Bind the certified transition to actual retained outputs
SCIF, PEAR [11, 12]	Inclusion or verifiable aggregation under their trust models	Use the external accepted set before first exposure
Stateful aggregation, PCD [14, 21]	Private aggregate state or evidence of computation history	Authorize the current per-identity head and atomic advancement
PrivStats, ZKLP [1, 22]	Upload accountability or hidden location predicates	Connect accepted history, retained output and release membership
SecAgg, Flamingo [23, 24]	Secure aggregation, dropout or multiple rounds	Preserve a complete prelocked cohort at first exposure
DP heatmaps, Prochlo [25, 26]	Output privacy or shuffled analytics	Supply an admitted-history input to the chosen release mechanism
SHTPC	Accepted-history closure	P1–P4 with correct services and separate single-cohort privacy

7. Discussion and Conclusion

This section discusses the deployment tradeoffs and limits of SHTPC.

Deployment tradeoffs.. Periodic reporting makes a fixed release reference useful when complete cohort statistics take priority over immediate subset availability. One internally inconsistent body or vanished predeclared member can hold the cohort, while durable delivery handles later transport failures. This tradeoff suits applications that value a complete declared cohort or explicit refusal over immediate partial statistics. Input coverage and a common start affect whether a prelocked complete set is available. The measured Full session includes six history-initialization steps, whose cost can be weighed against subsequent window scheduling.

The compact relation uses native Histogram validity, while native processing still scans the vector in $O(D)$. Other input domains require their own identifying relation and privacy analysis.

Output privacy.. Public activity and contribution bits remain visible, and a Collector who knows every other owner's cell can infer the remaining contribution from an exact total. The stated single-cohort privacy model protects the additional protocol views. Differential privacy for the output and composition across multiple releases require separate mechanisms and analysis.

SHTPC connects hidden-state authorization to the contributions retained by native private aggregation and to complete-cohort first exposure. Masked moments, atomic admission and durable reservation connect these boundaries under the stated service and trust assumptions. Native attack and recovery tests exercise the interfaces, while component and lifecycle measurements characterize their costs and the input screen distinguishes individual legality from simultaneous complete input.

8. Artifact Availability

The repository at https://anonymous.4open.science/r/semantic-closure-artifact-2300/updates/2026-10-29/ contains the manuscript, component sources and evidence map. Its proof/README.md maps the numbered G4b proof documents to the statements here. G4b 02 is proof/02-groth16-extraction-and-budget.md. G4b 03, 04 and 07 contain the native query ledger, full-view online simulation and ordered composition hybrids. The directory also includes their explicit budget contracts and version provenance. These detailed arguments use the stated three-key source model. Raw trajectories follow their original licenses.

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Appendix A.
Interface, Policy and Composition Details

Interface and policy.. An independent acceptance rectangle with valid pairs for two primaries also contains their mismatched cross-spliced pair. Thus every rectangle contained in the same-primary relation has one primary. For complete legal and illegal views X_0, X_1, let $a_i = \Pr[V(X_i) = 1]$. Completeness and soundness errors satisfy $e_c + e_s = 1 - a_0 + a_1 \geq 1 - |a_0 - a_1|$, which is one for identical views.

For real-trace replay, local equirectangular projection uses radius 6,371,000 m and centers (116.4, 39.9) for T-Drive and GeoLife, (-8.61, 41.15) for Porto, and (12.4964, 41.9028) for Rome. Coordinates are shifted by 5,000 m and floored to integer meters. Points outside [0, 9999] in either coordinate are excluded from legal histories. The earliest observation per 300-second slot is used without gap filling. Pressure histories require eight consecutive occupied slots and, for Porto, one trip. The screen uses $K = 6$, a 40 m/s speed tier and a 3,300 m anchor cap. The four-corpus screen uses the same slots, grid and policy, with corpus-specific parsers and trip or trajectory boundaries. Rome uses a cleaned raw-point cache with valid coordinates, a city bounding box and duplicate timestamps removed. Its date labels follow the original naive wall-clock calendar. Each corpus uses three frozen dates. GeoLife and Rome dates were selected by observed actor coverage before legality screening.

The hidden ADWC relation bounds squared displacement from the accepted predecessor by $\tau_u^2 \Delta t_t^2$, and from the anchor $\mathbf{a}_t = \mathbf{x}_{\max(0,t-K)}$ by $\min\{\tau_u^2 (\Delta T_t^{(K)})^2, c_{\text{policy}}^2\}$, with speed tier τ_u, anchor cap c_{policy} and $\Delta T_t^{(K)} = K \Delta t_{\text{slot}}$. Enrollment proves a hidden genesis. The first $\max(K, 2)$ accepted steps advance history without aggregation eligibility, and the authority checks the proof's commitment references against actual accepted history. Accepted steps, slots, timestamps and revisions are distinct.

Swap two owners' positive cells in metadata-compatible legal worlds. Complete aggregates coincide. A batch containing one swapped owner has different aggregates, so the corrupted role and Collector distinguish when the honest share arrives. A subsequent publication refusal cannot erase it. This proves Proposition 3.

Registration stores full signed candidates without advancing history. Lock fixes P_w, R_w before private processing, and permits reference only actual committed predecessors and anchors. Commit atomically checks head and quota and records the successor, history and signed decision. This linearization point enables exactly the next transition in Definition 1 and preserves real-time precedence. Apply changes $N_{S,w}(r)$ from zero to one only for eligible commits, with duplicate retries as no-ops. Reserve checks the registration projections in Eq. (11) and all record links, then stores an immutable selector and sum before send. Recovery preserves reservations; pre-exposure abort retains partial commits and post-send failure retains the exposed snapshot. Induction gives Eq. (7) at every prefix, and native fidelity gives Eq. (13).

Appendix B.
Joint Privacy and Concrete Bounds

Trust and lifecycle.. Clients hold report-signing keys, admission holds enrollment and authorization keys, and each role holds input HPKE and receipt keys. The shared native verification key is uniform in $\{0,1\}^{128}$ and hidden from clients in the integrity game. Only the Collector holds the aggregate HPKE secret. Finalizer uses authenticated admission records, and Decoder uses the decryption result and public verification keys.

Registration is authenticated and confidential. Bodies, permits, commits and receipts authenticate owner, task, window and context. Inputs and sidecars use recipient-bound labeled HPKE. Shares use the Collector key with context and set checks. Collector-to-Decoder delivery authenticates the finalized-set certificate. Preparation uses durable journals and leases. Transport may delay, drop or replay messages. Privacy includes observer messages under μ.

Credential epochs bind signed context while identity and quota survive rotation. Revocation, post-compromise recovery and forward secrecy are outside the model. Corrupted role state is included, honest role secrets and backups are not. Durability requires retention through replay checks and recovery.

BabyJubJub Pedersen commitments [52] use canonical prime-order points, fixed public bases and uniform full-domain blinders. The hidden genesis and transition relations bind the true predecessor and K-anchor. With native field $q = 2^{128} - 7 \cdot 2^{66} + 1$, canonical $r_A, r_R, T \in [0, q-1]$ and $x \in [0, D]$, where $D < q$, give $0 \leq x + r_A + r_R < 3q$. The carry $\chi \in \{0,1,2\}$ therefore makes $x + r_A + r_R = T + \chi q$ an integer equality. Both sides are below $3q < 2^{130} < p_{\text{proof}}$, where the BN254 scalar modulus satisfies $2^{253} < p_{\text{proof}} < 2^{254}$. Thus equality in the proof field cannot wrap. Successful counts fit a signed 64-bit integer, so reconstructed histograms do not wrap modulo q.

Lemma 3 (Origin isolation)**.** *Outside authentication, digest and origin-allocation failures, a protected ciphertext enters private processing only in its registered source body.*

Proof. Allocation binds the identifier to its owner before publication. The complete body, encryption labels, installed lock and process permit fix the input before decryption. A transplant changes authenticated bytes or requires an already allocated identifier. Exact retries reuse the source and durable native state. □

Fix finite budgets before the challenge. All counts include honest construction, rejected inputs, offline queries, retries and rolled-back work. For the three-key basis of Lemma 2, let $S = S_g + S_t + S_l$ count fresh external real or simulated source proofs and $V = V_g + V_t + V_l$ count verifications under the three keys. The $Q_{G_1}, Q_{G_2}, Q_{G_T}, Q_{\text{pair}}$

TABLE 9. ADJACENT HYBRIDS AND THEIR LOSS; N COUNTS HONEST NATIVE SOURCES AND Q_i THE SEVEN STREAM BUDGETS.

Step	Change and simulator knowledge	Loss
1	Nonce deduplication over public unprepared sources	$N^2/2^{128}$
2	Initial usage, seed and binder programming	$N[Q_4 + 2(Q_3 + Q_6 + Q_7) + Q_5]/2^{128}$
3	Share and part programming with corrupt shares fixed	$N[Q_1 + Q_2 + 2Q_7]/2^{128}$
4	Prover-coin removal for unexposed seeds	$NQ_4/2^{128}$
5	Source simulation, full verifier, joint receipts, raw trace and first exposure	0
6	Strict-time truncation after $L + 128$ candidates	t_N

budgets count typed group-oracle outputs beyond the separately counted sources, imports and verifications. The counts I_1, I_2 cover source-group imports, and native XOF queries Q_i use a separate budget. The proof-field ledger is

$$\begin{aligned} h_1 &= 330974 + 2S + Q_{G_1} + I_1 \\ &\quad + 13V_g + 53V_t + 13V_l, \\ h_2 &= 80986 + S + Q_{G_2} + I_2 + V, \\ h_T &= 5 + Q_{\mathrm{pair}} + Q_{G_T} + 7V, \\ d_G &= 32769\tbinom{h_1}{2} + 16383\tbinom{h_2}{2} + 49152\tbinom{h_T}{2}, \\ g_{\mathrm{coll}} &= \min\{1, d_G/(p_{\mathrm{proof}} - 16385)\}, \\ g_{\mathrm{aff}} &= \min\{1, S(3p_{\mathrm{proof}} - 2)/p_{\mathrm{proof}}^2\}, \\ g &= \min\{1, g_{\mathrm{setup}} + g_{\mathrm{coll}} + g_{\mathrm{aff}}\}. \end{aligned} \tag{16}$$

The h values bound group handles at one hybrid endpoint. Sampling directly from the specified good setup domain gives $g_{\mathrm{setup}} = 0$. Unknown trapdoors and independent setup entropy remain premises. The affine term is a conservative additional charge that can overlap the collision event. The fixed source inventory and derivation appear in artifact document G4b 02, Sections 2–7 (Section 8).

Let Q_i count native stream queries by domain, in order measurement, proof share, joint randomness, prover randomness, query randomness, joint seed and joint part. Put $Q = \sum_{i=1}^{7} Q_i$, and let P_{att} count distinct attack preparation inputs. The actual public key tag is a 96-bit prefix of the verification-key hash. For G_5 external candidate-key queries, let K be the uniform 128-bit key and $Z = f(K)$ its tag. On the no-hit branch, each tag value determines at most G_5 candidate keys E_z. Averaging over tags gives

$$\Pr[K \in E_Z] = 2^{-128} \sum_z |f^{-1}(z) \cap E_z| \le 2^{96} G_5/2^{128}.$$

Thus $\kappa_{\mathrm{tag}} \le \min\{1, G_5/2^{32}\}$, without assuming uniform tag fibers. The factor two below charges the two secret-key replacement hybrids, each with one such first-hit event. An additional SHA256 one-wayness assumption on uniform 128-bit inputs gives the alternative bound $\kappa_{\mathrm{tag}} \le a_{\mathrm{OW128}}$. Scanning all queried keys gives that reduction with one additional hash per candidate. Collision resistance alone does not supply this assumption. The native validity term is

$$\begin{aligned} v_N = \min\big\{1, 10256(Q + P_{\mathrm{att}})/q + 2\kappa_{\mathrm{tag}} \\ + (2P_{\mathrm{att}} + Q)^2/2^{128}\big\}. \end{aligned} \tag{17}$$

Here $10256 = 10002 + 254$. For 10,001 bins the invalid-input identity has error at most $(10001 + 1)/q$, while a false gadget polynomial of degree at most 254 contributes $254/q$. The root-abort branch remains in the experiment. This term concerns invalid-input acceptance. Privacy against a corrupted role, which already knows the verification key, has a separate nonce-based bound below.

Definition 7 (Exposure equivalence)**.** Using the declared indices of Definition 6, legal histories with common public structure μ are exposure-equivalent for R_w when

$$\sum_{i \in R_w} \mathbf{e}_{y_i^{(0)}} = \sum_{i \in R_w} \mathbf{e}_{y_i^{(1)}}. \tag{18}$$

Authority-only has no aggregate query and does not require this equality.

Proposition 4 (Effect of the exposure family)**.** *A fixed full cohort permits histogram-preserving reassignment. If every size-k subset can instead be exposed, $1 \le k < n$, equal sums on all such subsets force identical per-record native indices.*

Proof. Set $d_i = \mathbf{e}_{y_i^{(0)}} - \mathbf{e}_{y_i^{(1)}}$. Exchanging i, j in a size-k subset gives $d_i = d_j$. A size-k sum then gives $kd_i = 0$, hence $d_i = 0$ over integers. □

Full-domain blinding lets every hidden state point represent any fixed legal reference history. History-node aliases reuse the same point. This is language membership for the proof hybrid, not an extraction algorithm. Equation (14) gives the joint receipt law $(T, m_h) = (U, U - F(v_c^{\mathrm{orig}}) - r_c)$. For authority-only, uniform T, U give $(T, m_A, m_R) = (T, U, T - U)$.

The native model is a domain-separated programmable random oracle indexed by usage, seed, complete binders and raw offset. Repeated descriptors reuse rejected candidates and unconsumed prefetch too. Applying these bounds to the concrete TurboSHAKE128 XOF requires this idealization beyond XOF pseudorandomness. The native kernel specializes the two-party, single-proof analysis in the 2024-09-25 revised full version of [28] to the pinned Histogram interface. The following bounds and simulator account for this interface and its differences rather than directly copying the old bound.

Lemma 4 (Concrete Histogram preparation view)**.** *The two-role, single-proof 10,001-bin Histogram has a 455-element proof, 202-element verifier and 128 interpolation points*

Ω. *For valid inputs and query $t \notin \Omega$, its full verifier is $(0, W_0, \ldots, W_{199}, \sum_{j=0}^{99} W_{2j} W_{2j+1})$, with independent uniform W_i. Queries in Ω take the same root-abort branch in both worlds.*

Proof. Each valid-input wire has nonzero random-value coefficient $(t^{128} - 1)/(128(t-1))$. Its affine inverse makes evaluations independent uniform W_i, determining the gadget coordinate. For $t \in \Omega$ both worlds abort identically. □

Theorem 3 (Concrete two-party native privacy)**.** *In this programmable-oracle model, let N count honest native source reports, including unprepared reports. A single-world native stream simulation has loss at most*

$$Q_* = Q_1 + Q_2 + Q_5 + 2(Q_3 + Q_4 + Q_6) + 4Q_7,$$
$$b_N = \min\{1, (N^2 + NQ_*)/2^{128}\}. \tag{19}$$

Before exposure the simulator uses no aggregate query. At exposure it uses one complete-cohort query under Eq. (18)*.*

Proof sketch. Couple corrupt inputs and program fresh domain-separated streams. Nonce collisions and prior queries give Eq. (19). For either corrupt role, fix its original measurement/proof shares, key and nonce. Sample full verifier Σ from the lemma, compute the corrupt original distributed query z_c^{orig}, and set $z_h = \Sigma - z_c^{\mathrm{orig}}$. Each active response z_c' undergoes $\mathsf{Decide}(z_h + z_c')$ and seed-equality checks. Root queries reuse the abort branch. Helper exact retries reuse journaled bytes. Leader uncommitted recomputations use the same Σ, z_h and original state for changing responses, committing at most once. Verdicts and modeled times follow this state without real-verdict/time oracles. Recomputed work remains budgeted. G4b 03, Sections 3–7, and 04, Sections 2–7, give the descriptors and full online hybrids in the artifact (Section 8). □

Theorem 4 (Joint privacy of admission and native release)**.** *Use the privacy experiment of Definition 6, the source and programmable-oracle models above, and their primitive bounds. Let X denote authority-only, active-role-and-Collector, or that aggregation view with the authority's business view. Let g be the bound in Eq.* (16)*, ϵ_H the full-digest SHA256 collision advantage over all hash queries and bytes, a_X the sum of signature EUF-CMA advantages for correct-party keys unknown to the observer, and c_B the five-purpose Pedersen binding advantage. Let N_{id} count honest identifier draws, including failed reservations, and G_{id} count adversarial guesses before publication. With C_X hidden HPKE encapsulations, let H_X sum their labeled IND-CCA advantages and set*

$$B_X = \min\{1, g + \epsilon_H + a_X + c_B + N_{\mathrm{id}} G_{\mathrm{id}}/2^{128}\}, \tag{20}$$
$$e_X = \min\{1, 2(C_X + 10)B_X + H_X + b_N + t_N\}, \tag{21}$$
$$\mathrm{Adv}_X \le \min\{1, 2e_X\}, \tag{22}$$

where $t_N \le \min\{1, 5N\binom{10129}{129}\rho^{129}\}$ and $\rho = (2^{128} - q)/2^{128}$. Authority-only needs no aggregate equality. The other two views require Eq. (18)*.*

Hybrid outline. Rows 1–4 of Table 9 sum to the bound in Eq. (19). The C_X labeled HPKE hops and ten macro hops each have two endpoints, each charged B_X. Their endpoint charges give $2(C_X + 10)B_X$, while H_X adds one IND-CCA advantage per hidden encapsulation. The ten macro hops cover source proofs, nonce deduplication, initial oracle programming, shares and parts, prover coins, the full verifier, points and receipts, raw traces, business and exposure processing, and truncation. Artifact document G4b 07, Section 5, gives the ordered ledger (Section 8). The five field streams are measurement, proof share, joint, prover and query randomness. Seed and part streams output bytes. A retry preserves native randomness and original-source state, including the report, key, nonce and original shares. Active responses and verdicts can vary. Truncation after $L + 128$ candidates requires 129 rejections to fail. A union bound over $5N$ streams with $L \le 10001$ gives t_N. Repeated descriptors reuse their stream. The two histories share one ideal view, so their distances add. □

For aggregation views, $C_X \le 2N$ counts one hidden native input and one sidecar for the honest role per honest source. Authority-only has both roles' inputs and sidecars, so $C_X \le 4N + E_{\mathrm{enc}}$, where E_{enc} counts fresh aggregate encapsulations visible to the authority, including re-encrypted retries. Each term in H_X uses one challenge and its recipient-key-specific time, permitted decryption-query count and byte budgets. Primitive time budgets include the reduction's work. For integrity, the corresponding $X = C$ route bound is B_C. Malicious sources use authorized decryption and real native execution. Equation (15) subtracts their declared indices and adds their actual honest-role outputs.

Lemma 5 (Export refinement)**.** *On the legal reachable install domain, the optimized export transaction accepts the same reservations and payload bindings as the full-scan transaction.*

Proof. Unique unsigned-decimal window text selects the same reference row. For integer-only records, fixed JSON ordering/escaping and unique integer text make canonical and parsed equality equivalent. Floats are excluded. Other records use parsed comparison. The projection retains body, window, record and lock identifiers and reads complete records and the lock certificate from the reference. Both transactions check the same freeze, lock, count, commit, eligible retained-link, consumed membership, frozen-set, reservation and payload conditions under one control lock and repeatable-read snapshot. Locking one complete P keeps the reference consistent. Install rejects rows outside this domain. □